

Human Computer Interaction on the Web

Splash: A thermal assistant for your drinks

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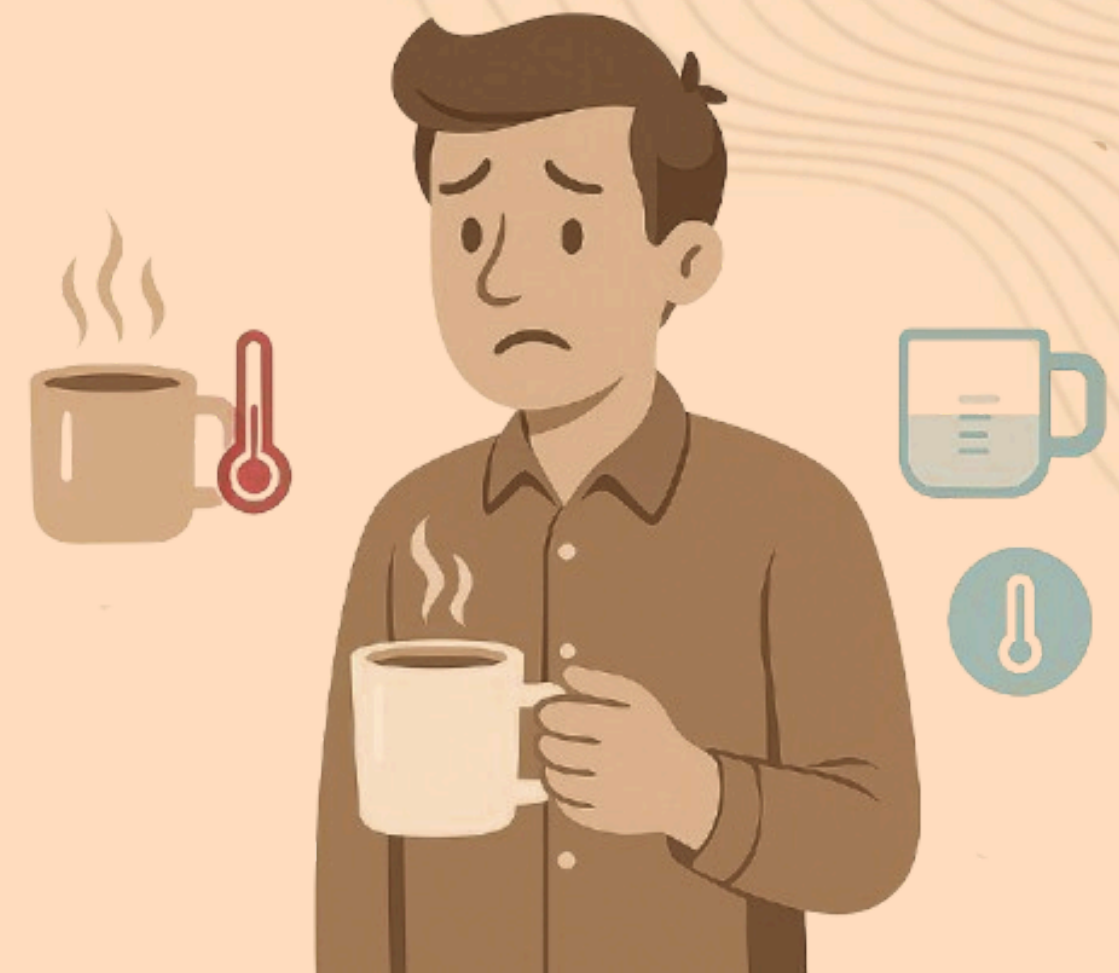


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THE PROBLEM

Drinks cool down or remain too hot without the user being aware of it:

- **Dynamic context:** home, work, study.
- **Lack of feedback:** the user never knows when it is the ideal time to drink.
- **Fragmented attention:** need for discreet, effective, and accessible notifications.



THE IDEA: SPLASH

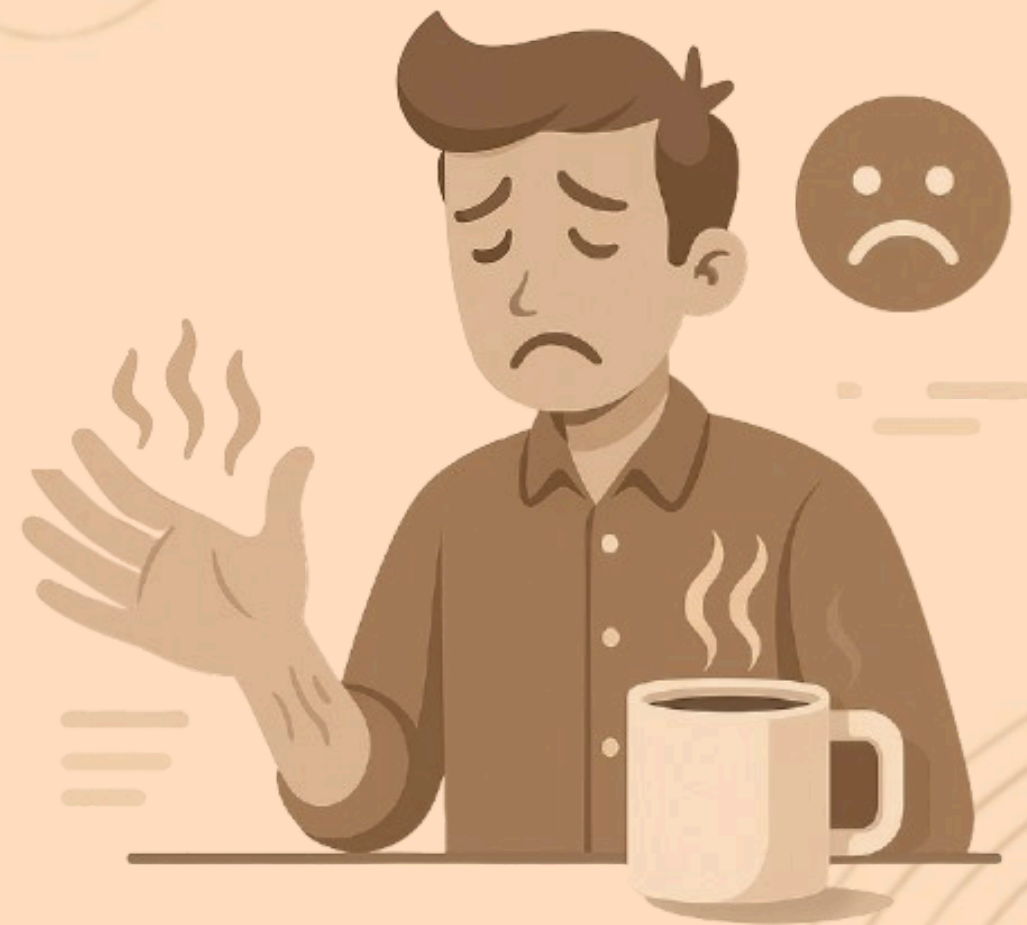


Splash is an integrated system based on:

- **Tangible interaction:** the user places a cup with an NFC chip on the surface.
- **Passive monitoring:** temperature controlled in real time.
- **Multimodal feedback:** display, vibration, mobile app.

→ Everything revolves around a simple, responsive, environmental user experience.

CONTEXT OF USE AND SCENARIOS



Most common situations:

- Drink too hot: risk of burns.
- At home or in the office, while busy: drink gets cold.
- Infants: milk at controlled temperature.

Target users

- Workers
- Parents
- Students
- etc...

SYSTEM WORKFLOW



The user places the cup
→ identification via NFC.



Before it cools down too
much, a new alert is sent.



The system reads the
temperature every second.



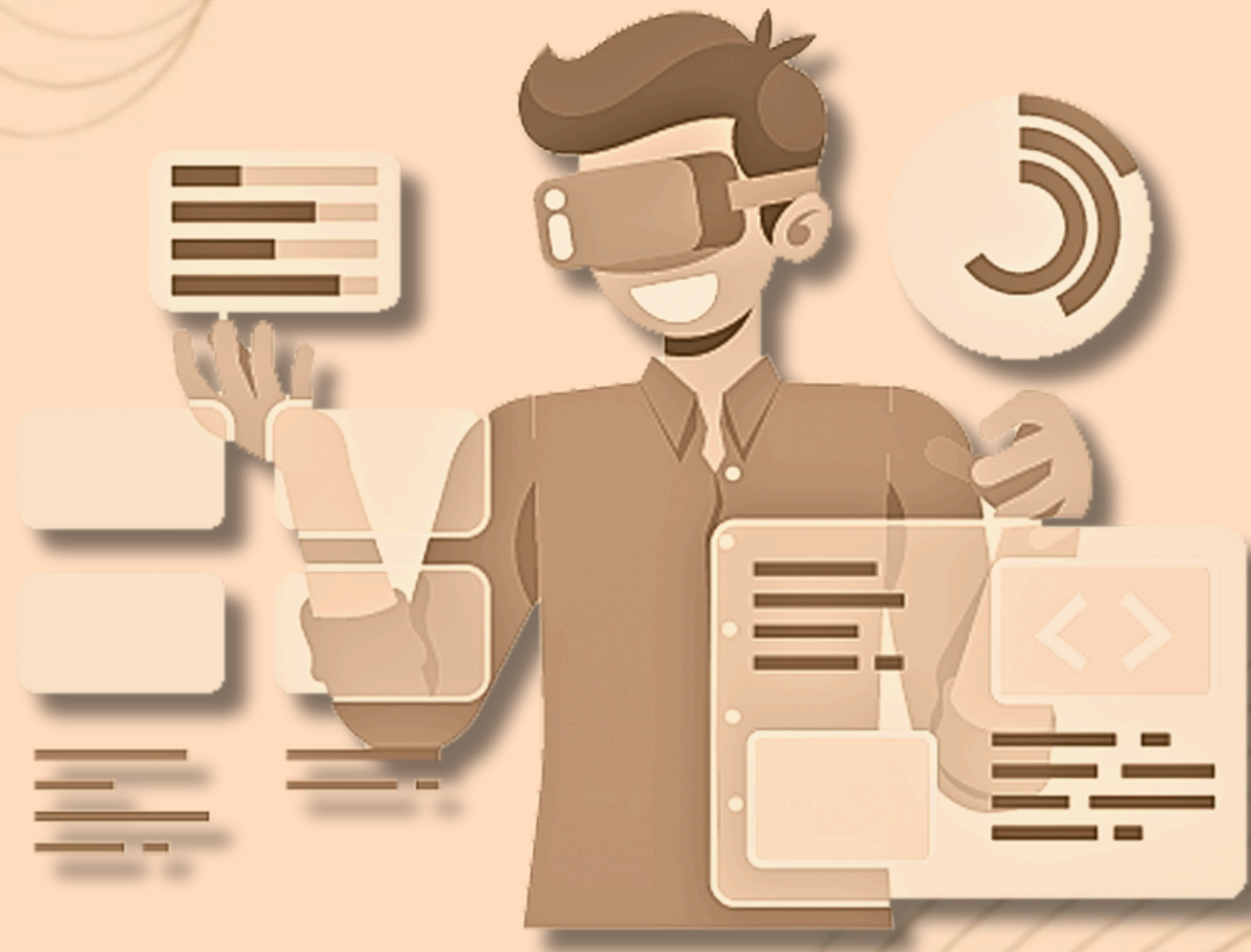
The user can remove the
cup.



If it is within the optimal
range: it vibrates, changes
color, and sends a notification.

**A responsive and simple loop,
adapted to the user's attention.**

TYPES OF INTERACTION



- **Tangible Interaction:** the cup with NFC activates recognition.
- **Audio Feedback:** vibration when the temperature is optimal.
- **Visual Feedback:** display changes color.
- **Asynchronous feedback:** notification via app → distributed multimodality.
- **Customization of settings via app.**

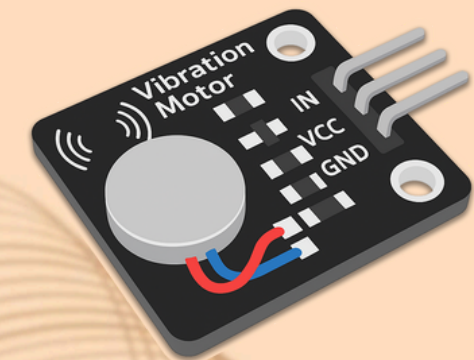
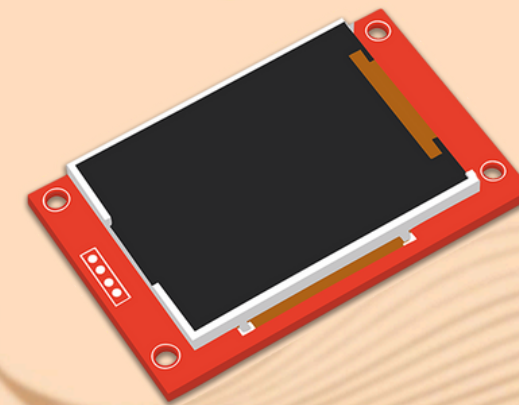
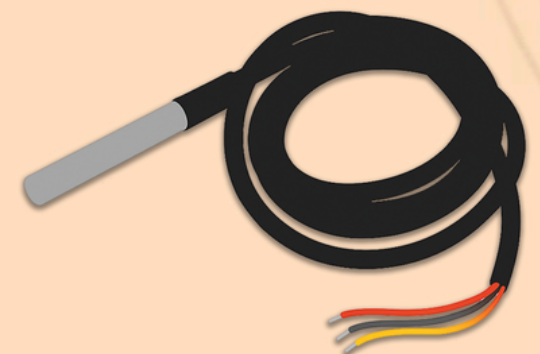
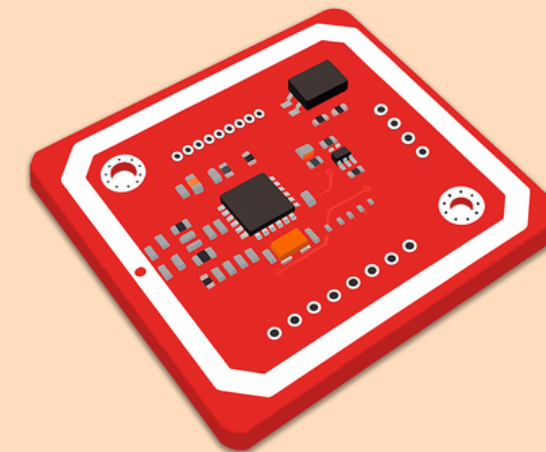
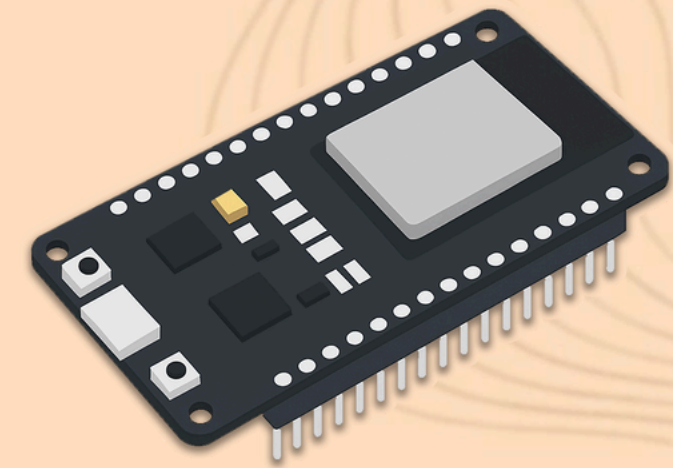
SYSTEM ARCHITECTURE

Hardware:

- **ESP32:** brain of the system, manages sensors, actuators and communication.
- **Temperature Sensor DS18B20:** Measures the temperature of the drink by contact with the cup.
- **NFC reader + NFC tag:** to detect the type of the drink.

Output:

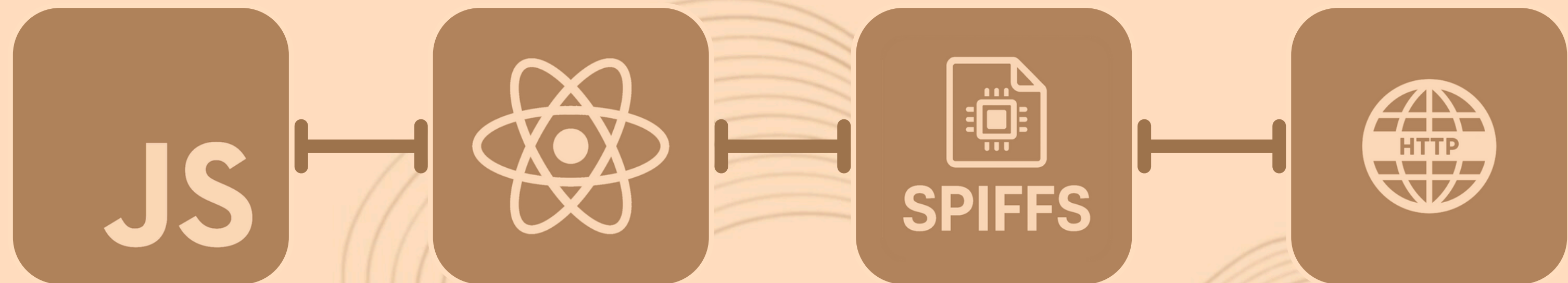
- **Vibrating motor:** generates a discrete sound when the optimal temperature is reached.
- **TFT screen:** shows temperature in real-time and gives feedback on drink status.



SYSTEM ARCHITECTURE

Software & Communication:

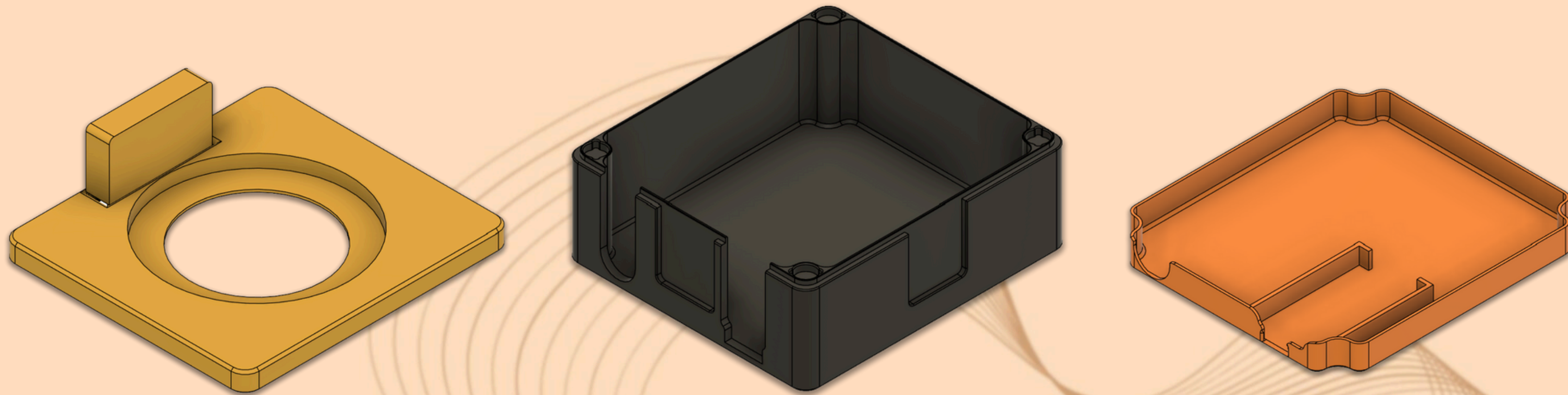
- **App:** Mobile application developed with React Native.
- **WebSocket + HTTP API:** Continuous communication with the app.
- **SPIFFS:** Custom storage of the "ideal temperature."



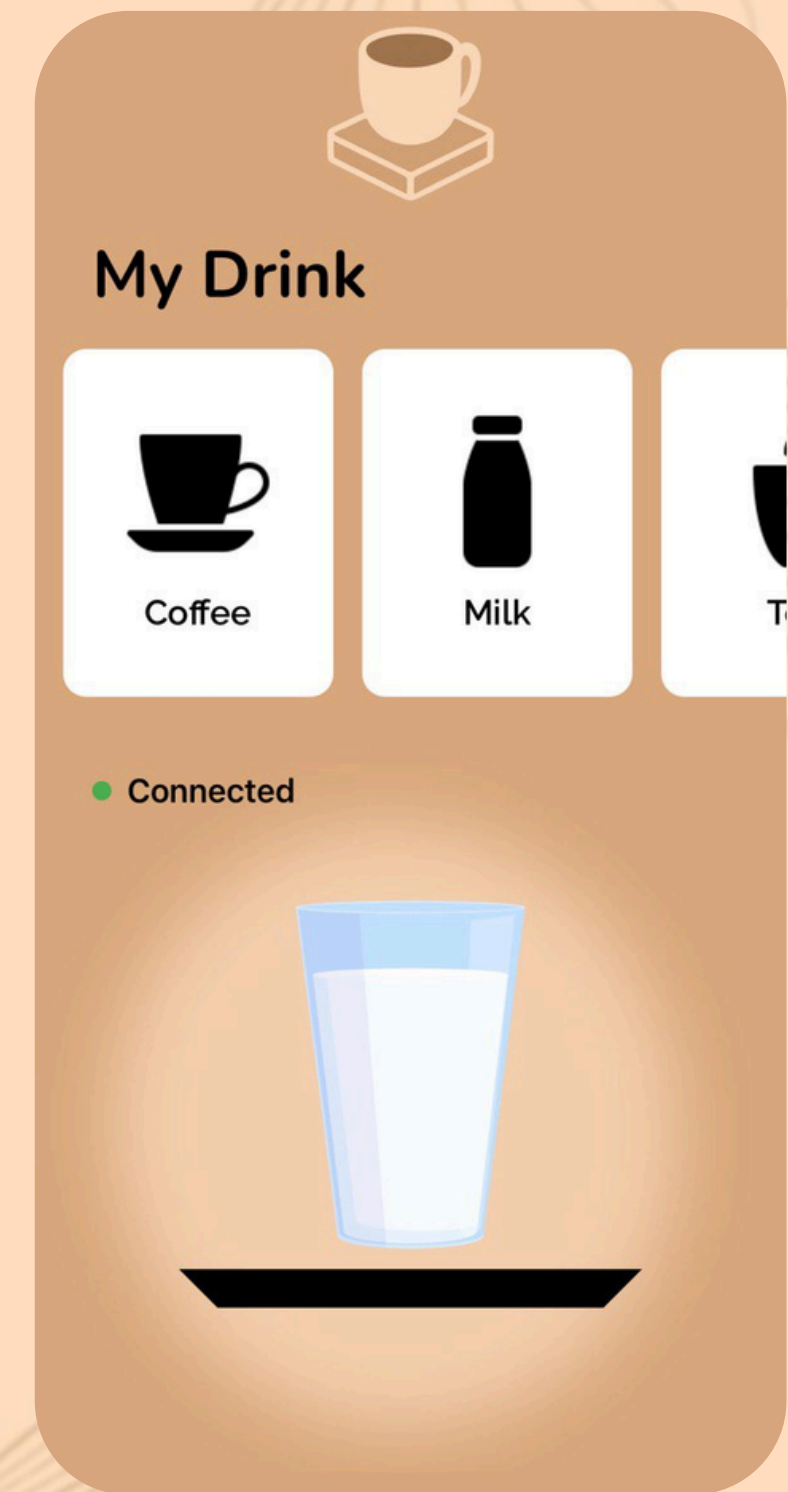
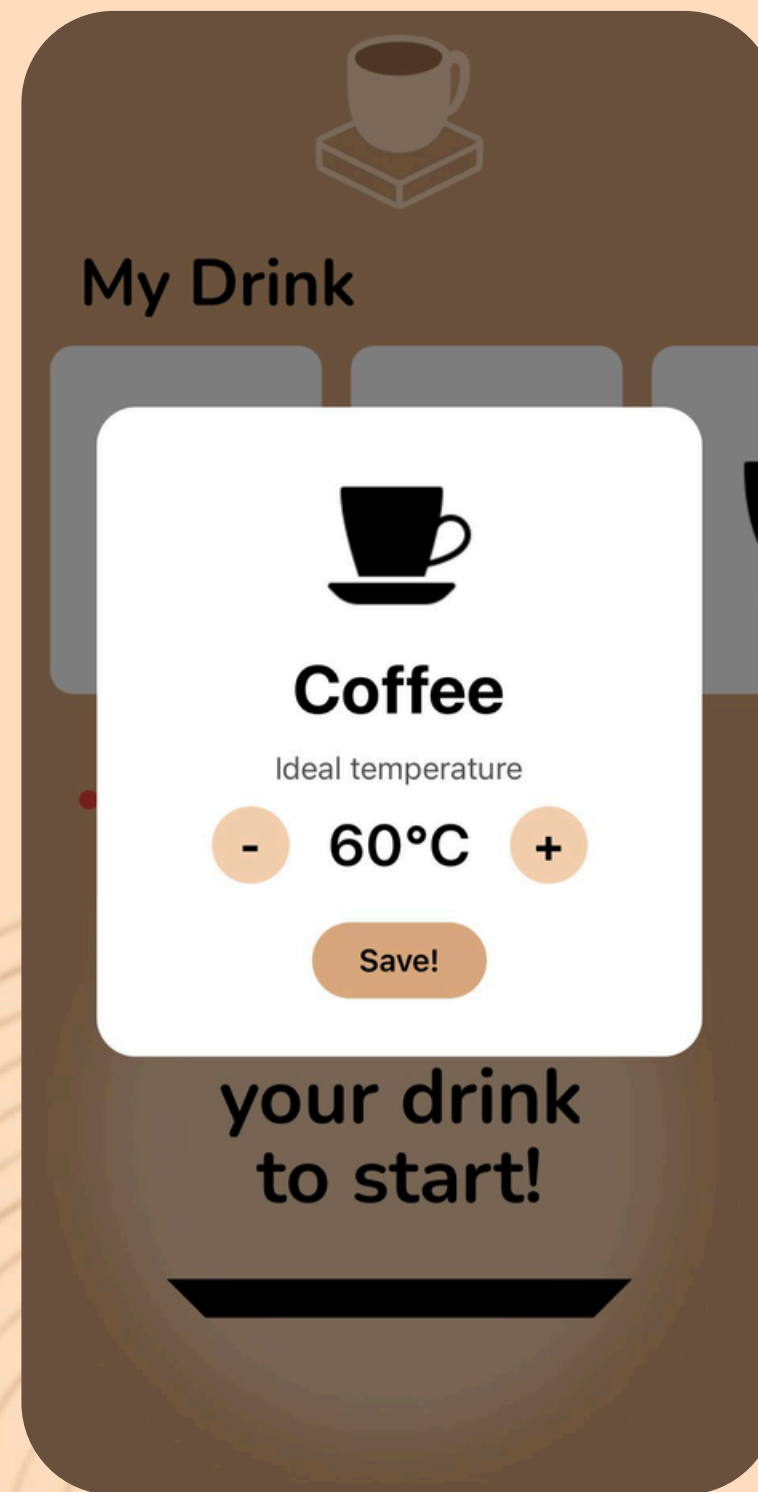
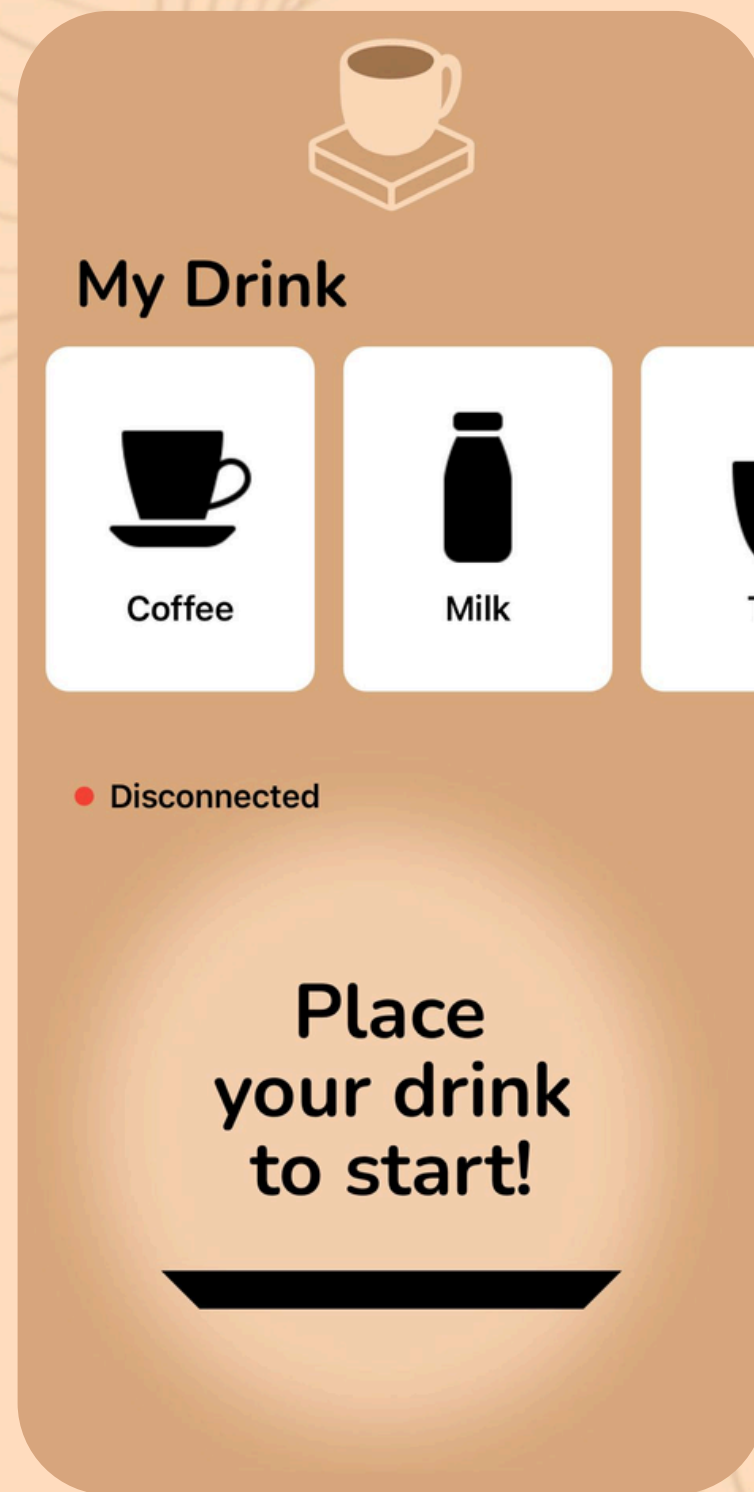
3D DESIGN & PROTOTYPING

We designed the case in 3D to integrate all components:

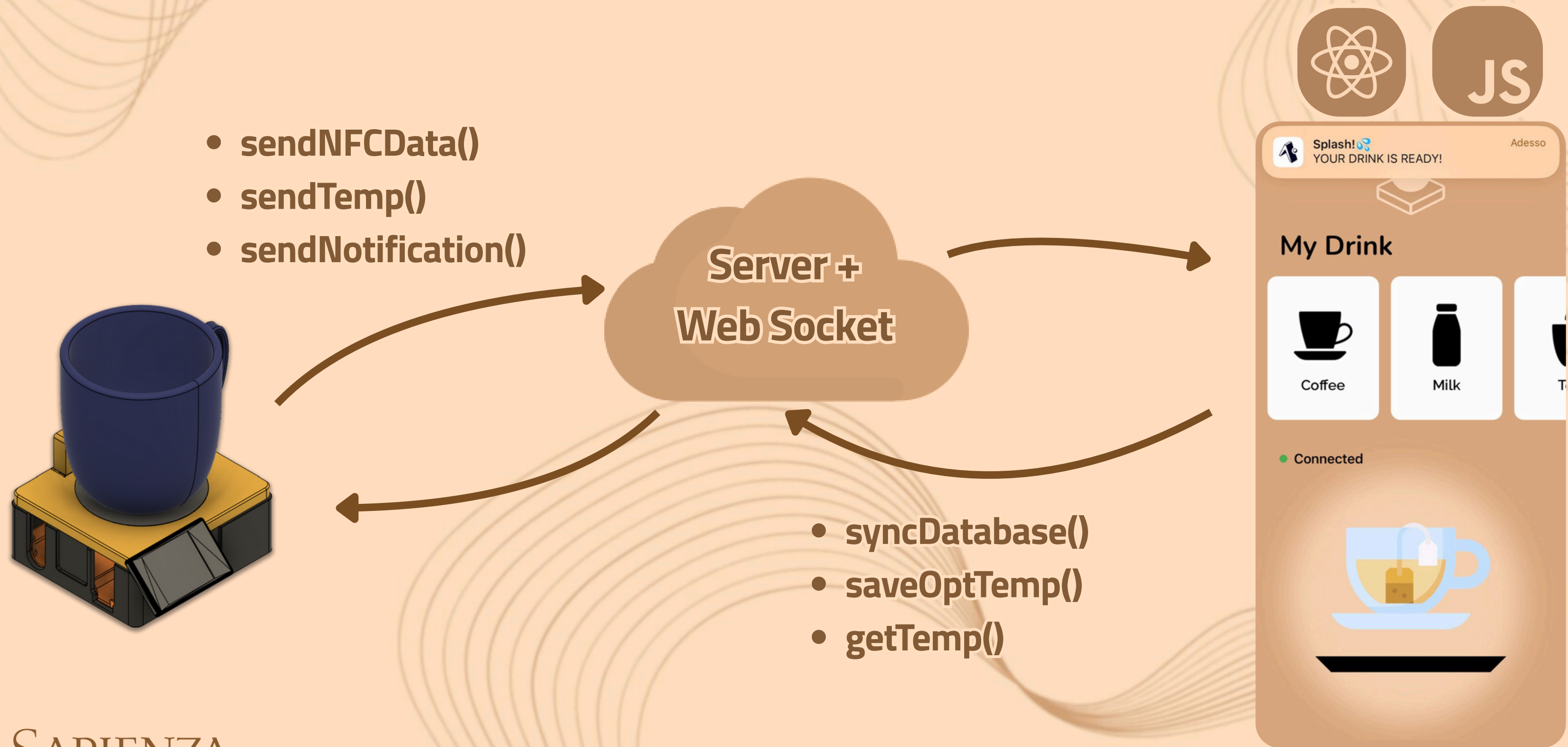
- The base houses the ESP32, temperature sensor, motor, and NFC reader.
- An add-on component allows for screen installation.
- The design is compact, 3D printable, and modular.



APP & UX DESIGN



SYSTEM OVERVIEW



CONCLUSIONS

Conclusions:

- Working temperature sensing system, easy and intuitive to use.
- Mobile app for customization according to user's taste and notification system.

Possible future developments:

- Investigate a more appropriate sound feedback system.
- Add a system to maintain the cup within the optimal temperature range.
- Add further customization options via the app (e.g., adding new drink cards).



**THANK YOU FOR YOUR
ATTENTION**



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